

BRIEFING SLIDES

$$\sum_{k=1}^n$$

$$\binom{n}{2}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

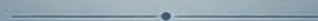


$$c^2 = a^2 + b^2$$

$$y = \sin x$$

MATHEMATICS

Think. Understand. Solve.



$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



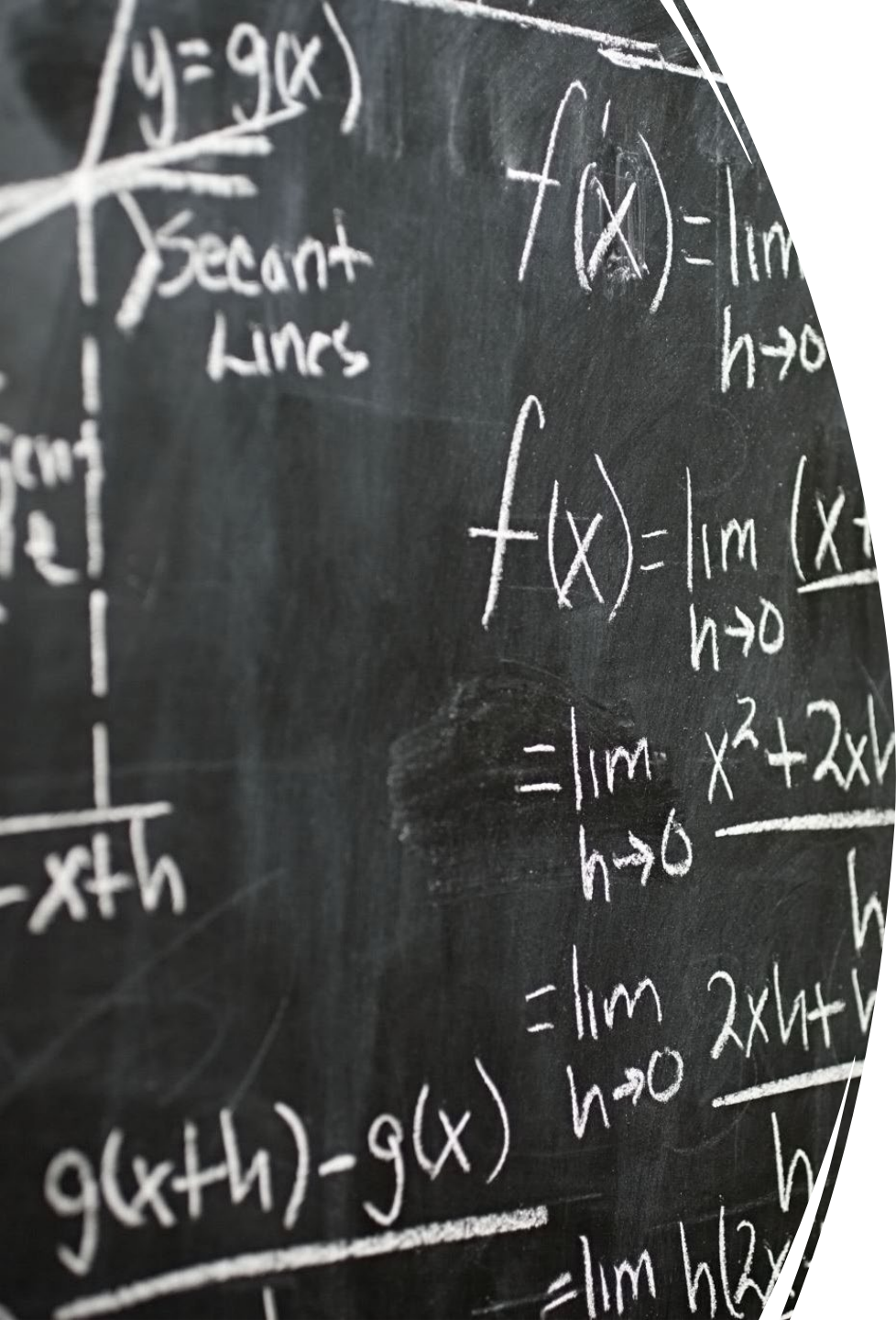
$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

University Preparatory Programme Mathematics

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

September 2026





Welcome!

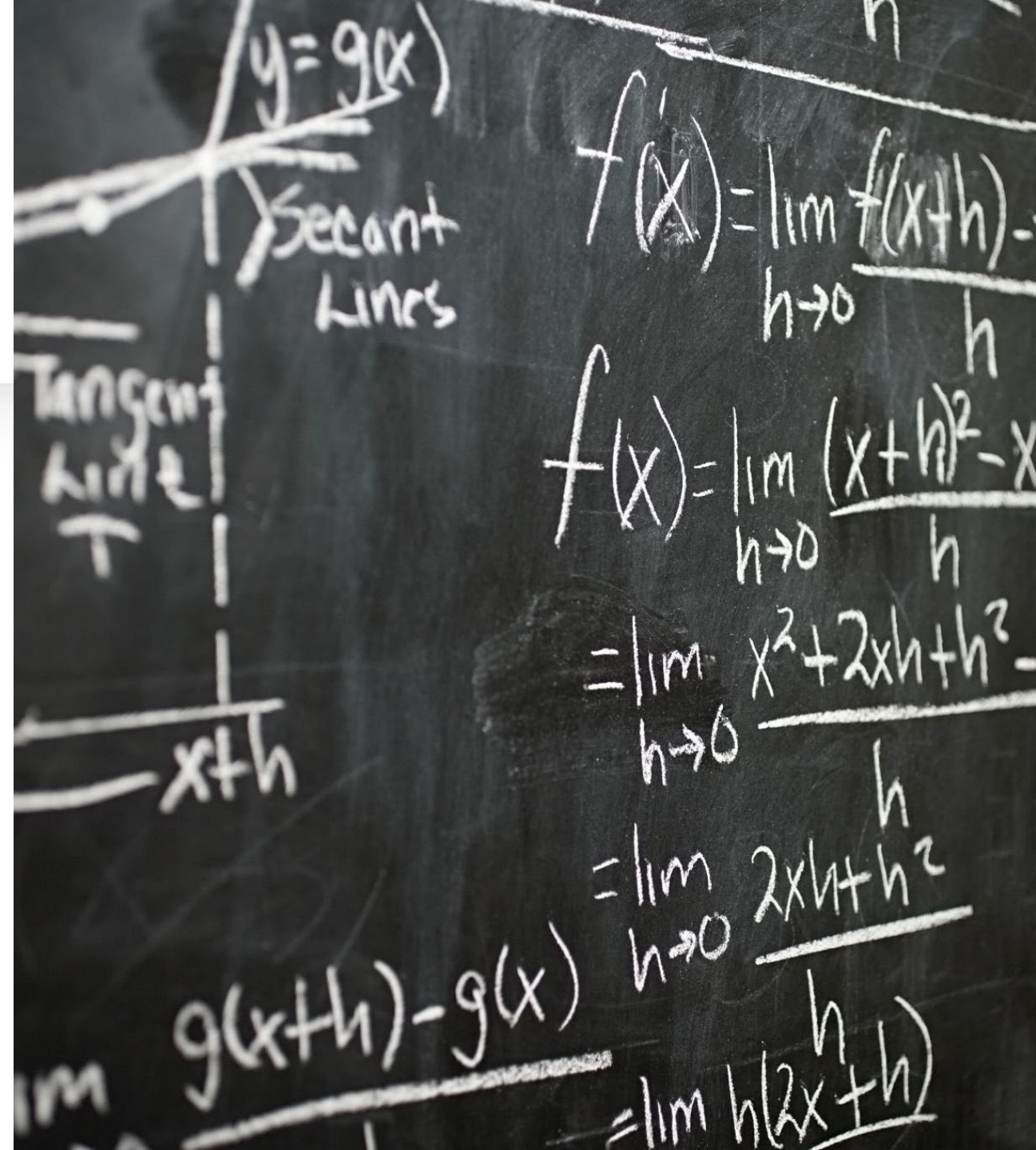
Synopsis

This course will provide student-participants with an in-depth knowledge of calculus. Students will be introduced to numerical methods and the concepts of improper integral, infinite series, 3D vectors, and techniques to solve ordinary differential equations. They will also learn topics from calculus of multi-variable functions such as partial differentiation and multiple integrals.

Objectives

Upon completion of this course, participants will be equipped with the knowledge and skills to:

1. Apply numerical methods to solve simple algebra and calculus problems.
2. Apply Limit techniques to calculate improper integrals and test the convergence of infinite series.
3. Solve problems involving 3D vectors.
4. Calculate the derivatives and integrals of two-variable functions.
5. Solve Ordinary Differential Equations (ODEs).



What are we doing Today?

- A. Admin matters
- B. Overview of programme
- C. Topic 1 Numerical Methods
 - 1. Approximate a root of a function using Newton-Raphson method
 - 2. Approximate an integral using Trapezoidal Rule and Simpson's Rule
 - 3. Approximate a solution of an initial value problem using Euler or 2nd Order Taylor method

LMS for UPPM: [Cousebox.ai](https://cousebox.ai)

 **Coursebox**

HOMEABOUTCOURSESCONTACTLOG IN

Request Access



☐ * I have read and agree to the terms of the User Agreement and Privacy Policy .

Register and Request

Already have an account? [Log in](#)

UPP Mathematics - Learning Hub

This course will provide student-participants with an in-depth knowledge of calculus. Students will be introduced to numerical methods and the concepts of improper integral, infinite series, 3D vectors, and techniques to solve ordinary differential equations. They will also learn topics from calculus of multi-variable functions such as partial differentiation and multiple integrals.

Created by  Ajay Kumar

Course Objectives

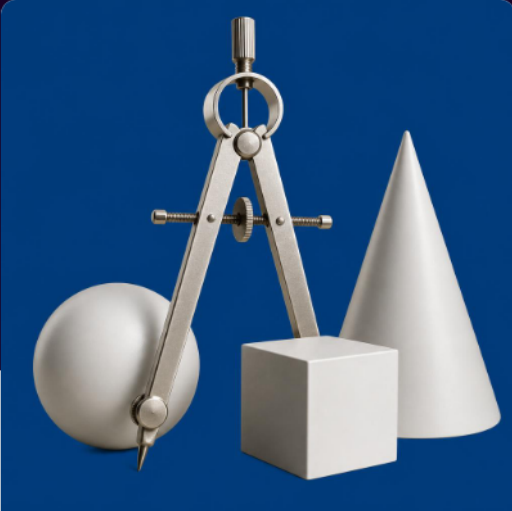
- Apply numerical methods to solve simple algebra and calculus problems.
- Apply Limit techniques to calculate improper integrals and test the convergence of infinite series.
- Solve problems involving 3D vectors.
- Calculate the derivatives and integrals of two-variable functions.
- Solve Ordinary Differential Equations (ODEs).

Course Content

10 sections · 48 lessons · 3h 56m

1. Course Introduction

5 lessons ^



UPP Mathematics - Learning Hub
by Ajay Kumar

A. Admin matters

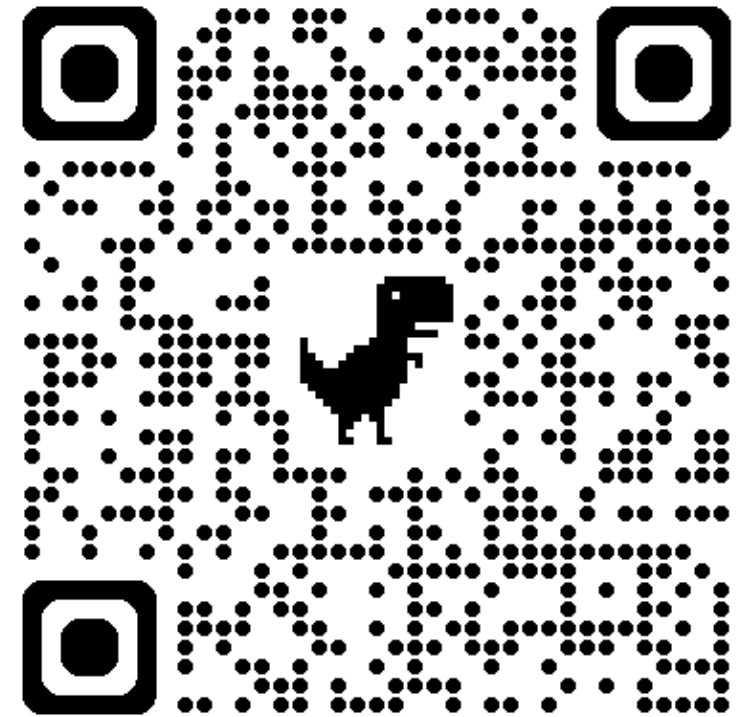
Students, you should have been invited via email and enrolled into the Coursebox.ai LMS already.

TP students – via TP school email

Other students – via the email you signed up for UPPM

Thereafter, you can access it @

- <https://my.coursebox.ai/courses/01a04124-05f4-7bc8-b826-b53a28d0c5c3/about>



B. Overview of programme

- All resources are hosted on the LMS
 - You may view/download/print the learning notes and other resources directly from the LMS

1. Course Introduction	5 lessons
2. Assignment	2 lessons
3. Ch.1: Numerical Foundations	2 lessons
4. Ch.2: Improper Integrals	9 lessons
5. Ch.3: Infinite Series	5 lessons
6. Ch.4: Vectors (SDL)	11 lessons
7. Ch.5: Partial Differentiation	4 lessons
8. Ch.6: Multiple Integrals	4 lessons
9. Ch.7: Ordinary Differentiation Equations	3 lessons
10. Mock / Sample Papers	3 lessons



UPP Mathematics - Learning Hub
by Ajay Kumar

Resume

B. Overview of programme

Date	Topic	Remarks
7 Sep 2026	Ch.1: Numerical Methods	Assignment is based on Ch. 1
9, 10 Sep 2026	Ch. 2: Improper Integrals & Ch.3: Infinite Series	Ch. 4: Vectors (Self-Directed Learning) *To be completed BEFORE Partial Differentiation
14, 16, 17 Sep 2026	Ch.5: Partial Differentiation	
21 Sep 2026	Revision / TEST 1 (30%)	Test 1: Chapters 2 – 5
23, 24, 28 Sep 2026	Ch. 6: Multiple Integrals	Assignment (30%) due on 25 Sep 2026
30 Sep & 1, 5 Oct 2026	Ch. 7: Ordinary Differential Equations (ODE)	
7 Oct 2026	Revision / TEST 2 (30%)	Test 2: Chapters 6 – 7

SDL on Vectors
e-Lectures
before Partial
Differentiation!

Course Title : University Preparatory Programme (UPP) - Mathematics

Intake No. : 2026 Sep (7 - 10 pm)

Classroom Venue: **Class 1: 10-6-14, Class 2: 10-6-15**

Time Table for UPP MATHEMATICS (7 Sep - 7 Oct 2026)

Subject/Time		MONDAY		TUESDAY		WEDNESDAY		THURSDAY		FRIDAY	
		7pm-10pm				7pm-10pm		7pm-10pm		7pm-10pm	
Wk	2026										
1	Sep	7	Numerical Methods	8		9	Improper Integral & Infinite Series	10	Infinite Series	11	
2	Sep	14	Partial Differentiation	15		16	Partial Differentiation	17	Partial Differentiation	18	
3	Sep	21	Revision / Test 1 (7:00 - 9.00 pm)	22		23	Multiple Integrals	24	Multiple Integrals	25	Assignment Due
4	Sep/Oct	28	Multiple Integrals	29		30	Ordinary Differential Equations (ODE)	1	ODE	2	
5	Oct	5	ODE	6		7	Revision / Test 2 (7:00 - 9.00 pm)	8		9	

Topics

Numerical Methods

Improper Integral & Infinite Series

Partial Differentiation

Multiple Integral

Ordinary Differential Equations (ODE)

Vectors: *Self-Direction Learning*

B. Overview of program

➤ Total of 4 assessments

1. Assignment (on Numerical Methods) = 30% - *Due on 25 September 2026*

2. Written Test 1 (on Improper Integral and Infinite Series, Vectors and Partial Differentiation, 90 minutes) = 30%

3. Written Test 2 (on Multiple Integral and ODE, 90 minutes) = 30%

4. Online mini quizzes (self-paced on Chapters 2,3,4,5,7 directly on LMS @ 2% each) = 10%

➤ **There will be NO makeup for missing the Written Tests and NO makeup for late comers. You must be physically present and on time for each respective Written Test.**

➤ **Students may use their physical calculator (any, but memory must be cleared), but strictly no e-calculator / calculator apps.**

Online mini quizzes (self-paced)

Exit course

Course

UPP Mathematic...

100%

5. Ch.3: Infinite Seri...

5.1 Learning Notes ✓

5.2 Solution ✓

5.3 Video Lecture (O...

5.4 Outline

5.5 Quiz

6. Ch.4: Vectors (S...

6.1 Learning Notes

6.2 Solution

6.3 Video Lecture 1

6.4 Video Lecture 2

6.5 Video Lecture 3

6.6 Video Lecture 4

6.7 Simulated Lecture 5

6.8 Animations

6.9 Vector Proofs

6.10 Outline

To clear this sub-topic, you must answer both questions in this quiz correctly.

This quiz is worth 2%.

Quiz

To pass the quiz you must answer 100% of the questions correctly.

Get Started

< Previous

Next >

Online mini quizzes (self-paced)

Online mini quizzes (10%)

- *To be all completed all via this LMS. From Ch. 2,3,4,5,7 only
@ 2% each*

All quizzes to be completed by last day of the programme
7 October 2026

Assignment – Section 1.6 *Instructions only first*

Exit course

Course

UPP Mathematic...

100%

1. Course Introducti...

1.1 Welcome

1.2 Schedule

1.3 Administrative Matt...

1.4 Formula Sheet

1.5 Calculus - RECAP

1.6 Assignment (Instru...

2. Ch.1: Numerical F...

2.1 Learning Notes

2.2 Outline

3. Ch.2: Improper In...

3.1 Learning Notes

3.2 Solution

3.3 Limits - Video Lect

Assignment

(Instructions only first)

Please read the instructions for the Assignment here carefully.

The Submission page for the Assignment is found proper under this LMS at Section 9: Assignment. Since this Coursebox.ai LMS follows a sequential section completion/progression for Learners, you'll need to complete the preceding sections (up to Section 8: Chapter 6 Multiple Integrals) before you can 'unlock' / access the Submission page.

Meanwhile, you can already start your Assignment after reading the instruction below.

Take note: Assignment is due **25 September 2026, 2359H**, and it constitutes **30% of your grade**.

1. This Assignment is inspired from the Exercises in **Chapter 1: Numerical Methods**.

Previous

Next

Assignment – Section 9: *Instructions + Submission*

Exit course

Course

UPP Mathematic...

100%

8.3 Outline

9. Assignment

9.1 Instructions

9.2 Submission

10. Ch.7: Ordinary D...

10.1 Learning Notes

10.2 Solution

10.3 Outline

10.4 Quiz

11. Mock Test 2

11.1 Intro

11.2 Mock Test 2

11.3 Mock Test 2 - Solu...

Assignment (Instructions)

Please read the instructions for the Assignment here carefully.

1. This Assignment is inspired from the Exercises in **Chapter 1: Numerical Methods**.
2. You are to complete the following 5 problems (refer to the attached image on the left).
3. Download the spreadsheet template UPP Assignment (Student Fullname).xlsx (refer to the attached file below) and complete each problem on a corresponding worksheet tab.

1. With the help of a spreadsheet app, use the Newton-Raphson method to approximate the root(s) of the function $f(x) = x^2 - x + \frac{1}{2}$ with accuracy of $\varepsilon = 10^{-6}$. You need to approximate all roots to obtain full credit. You are given that the root(s) lie in the interval $[-2, 2]$.
2. With the help of a spreadsheet app, use the Trapezoidal rule to approximate the area bounded by the curve $y = \frac{2x-1}{8+(2x-1)^2}$ and the lines $y=0$, $x=0$, and $x=1$ with 20 sub-intervals.
3. With the help of a spreadsheet app, use Simpson's rule to approximate the areas in Problem 2 with 10 sub-intervals.
4. With the help of a spreadsheet app, use the Euler method to approximate the solution of the IVP

< Previous

Next >

Assignment – Section 9: *Instruction + Submission*

The screenshot displays a course management system interface. On the left is a sidebar with a course outline. The main content area has a dark header with the title 'Assignment (Submission)' and a sub-header with submission instructions. Below this, there is a section for submission details and a list of instructions. A red circle highlights the '9.2 Submission' item in the sidebar. A small profile picture of a man is visible in the bottom right corner of the main content area.

Course Outline (Left Sidebar):

- Exit course
- Course
- UPP Mathematic...
- 100%
- 8.3 Outline
- ✓ 9. Assignment
- 9.1 Instructions
- 9.2 Submission
- ✓ 10. Ch.7: Ordinary D...
- 10.1 Learning Notes
- 10.2 Solution
- 10.3 Outline
- 10.4 Quiz
- ✓ 11. Mock Test 2
- 11.1 Intro
- 11.2 Mock Test 2
- 11.3 Mock Test 2 - Solu...

Assignment (Submission) Header:

Please refer to all the Submission instructions on this page carefully as it constitutes 30% of your grade.

Submission Instructions:

Submit your completed spreadsheet file **by 25 September 2026 [2359 hrs] at the bottom of this page.**

Instructions

- Click the **Upload** button to select your completed spreadsheet on your device

(Ensure you have saved your completed spreadsheet file with your Student Fullname indicated in the filename)
e.g. UPP Assignment (Ajay Kumar).xlsx

Navigation:

- < Previous
- Next >

Assignment (*Please take note!*)

- Instructions-only can already be accessed from Section 1.6 onwards. (i.e. first lesson)
- Submissions for Assignment is found proper under Section 9.2 (*since Coursebox.ai follows a sequential section completion / progression for Learners*), so you'll need to complete the preceding sections (up to Section 8: Chapter 6 Multiple Integrals) before you can 'unlock' / access the Submission page.
- Assignment is due 25 September 2026, 2359H, and it constitutes 30% of your grade.

Lastly... Attendance!



Do note that you need to be physically present for all classes for your attendance to be marked.



If you can't make it for lesson, there is **no need for a medical certificate** however you will be **marked as absent**.



This applies to both **Written Test 1 and Test 2**, **and zero mark will be awarded** for absence towards the respective tests.



Take care of yourself, prioritise attendance, and reach out if you need clarification.

Everything else can be found
on Coursebox.ai LMS...

Have a good programme ahead! 😊